

```
(format "False")) ; produces "True"
```

The *when* condition statement is a variant of the *if* statement written as a macro. Macros are an important feature of Lisp programming that allow you to extend the language. They are primarily code generators. If the 'when' condition is true, then it evaluates the then-forms. The syntax and an example are given below:

```
(when condition then-forms)
```

```
(when t
  (print "Abracadabra!")) ; produces "Abracadabra!"
```

Looping

You can use the *while* statement to perform looping in Emacs Lisp. The body of the 'while' loop is executed as long as the condition is true. An example that prints the numbers 0 to 4 is shown below:

```
(progn
  (setq i 0)
  (while (< i 5)
    (print i)
    (setq i (+ i 1))))
```

The *dolist* and *dotimes* macros can also be used for looping. The other approach is to use recursion. A literal definition in Emacs Lisp for computing the Fibonacci series is given below:

```
(defun fib (n)
  (cond ((= n 0) 0)
```

```
((= n 1) 1)
(t (+ (fib (- n 1))
      (fib (- n 2))))))
```

The *cond* control structure takes a series of clauses; each has a condition and a body-form. If any condition evaluates to true, the body-form is executed. Otherwise, it proceeds to the next clause.

An iterative version of the Fibonacci series is given below:

```
(defun fib (n)
  (fib-helper n 0 1))

(defun fib-helper (n a b)
  (if (= n 0)
      a
      (fib-helper (- n 1) b (+ a b))))

(fib 10) ; produces 55
```

The Emacs Lisp Cookbook is very handy (<https://www.emacswiki.org/emacs/ElispCookbook>).

You are encouraged to read 'An Introduction to Programming in Emacs Lisp' at <https://www.gnu.org/software/emacs/manual/eintr.html> and the Emacs Lisp Reference Manual at <https://www.gnu.org/software/emacs/manual/elisp.html>. 

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